Whisper Notes

Design Draft — Milestone 2

Frontend Systems · SoSe 2026 · Prof. Dr. Peter Braun

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Application Overview

Whisper Notes is a React web application that lets students upload a recorded lecture audio file and receive a text transcription generated by the open-source Whisper model. The interface is intentionally minimal: the primary users are exchange and international students who may be unfamiliar with the local language, so every screen should be self-explanatory and free of unnecessary controls.

The application consists of four views: **Home**, **Transcribe**, **Results**, and **About**. A persistent top navigation bar connects all views.

Main Screens

Login

The initial entry point of the application. Instead of a traditional landing page, users are presented with a simple authentication interface. Once the user successfully logs in, they are automatically redirected to the Transcribe view to begin their workflow.

Transcribe

The core view of the application. It contains an audio upload form with drag-and-drop support, format validation (.mp3, .wav, .m4a, .flac), optional language selection, and a submit button. During processing, a loading indicator replaces the button. From here, users can also navigate to their Shared Files. The wireframe is shown in Section 3.

Results

Displays the transcription text returned by the backend in a scrollable text area. Below the text area, two action buttons allow the user to copy the result to the clipboard or download it as a plain `.txt` file. A *New Transcription* button resets the form and navigates back to the Transcribe screen. The wireframe is shown in Section 3.

Shared Files

A view dedicated to managing the user's past work. It displays a list of previously transcribed and shared audio files. This screen allows users to quickly review past transcriptions or access shared files, with simple navigation to return to the main Transcribe view.

Wireframes and Mockups

The following mockups represent the structural layout of the application's primary screen states.

1. Login View

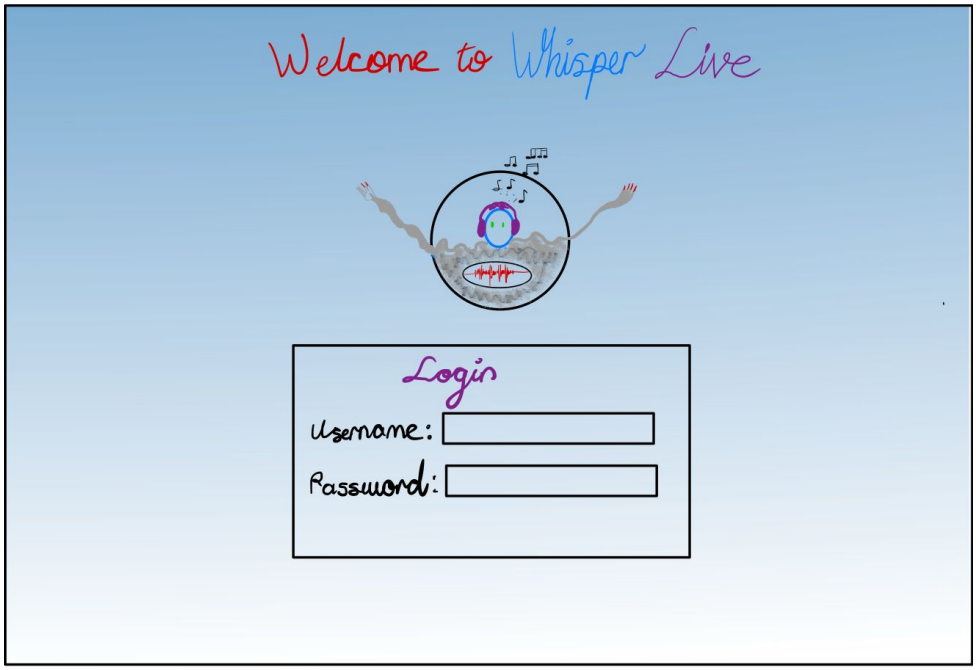


Figure 1: Login View Mockup

2. Transcribe View (Default)

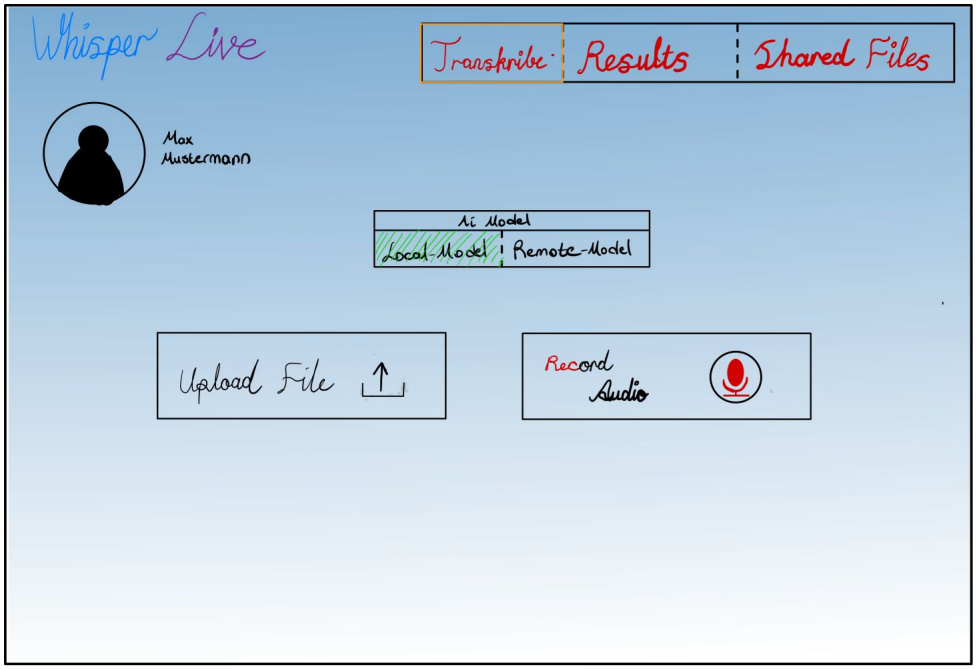


Figure 2: Transcribe View - Default Layout

3. Transcribe View (Upload File Selected)

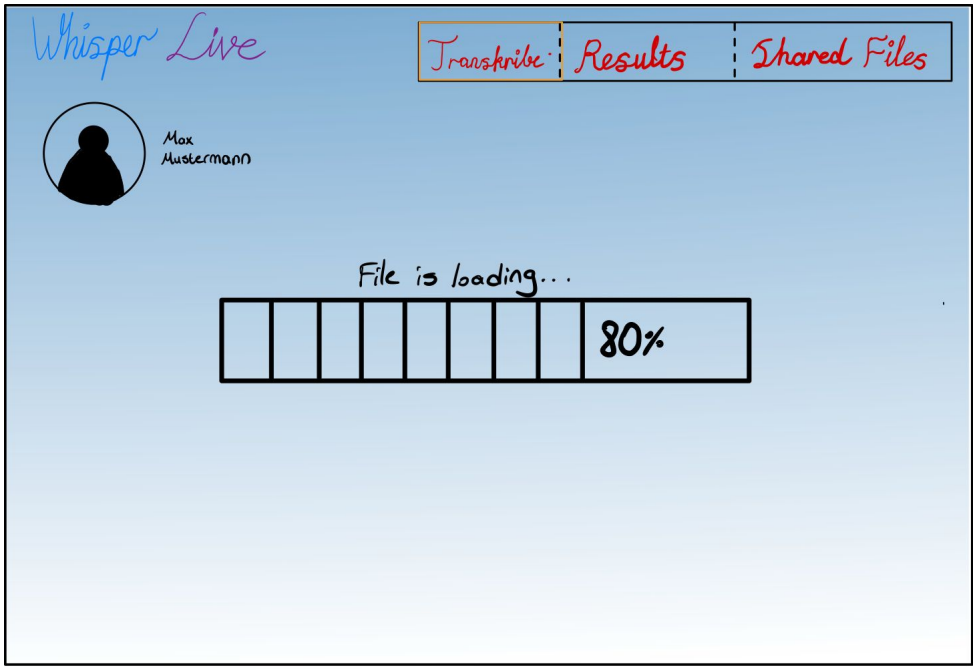


Figure 3: Transcribe View - File Upload State

4. Transcribe View (Record Audio Selected)

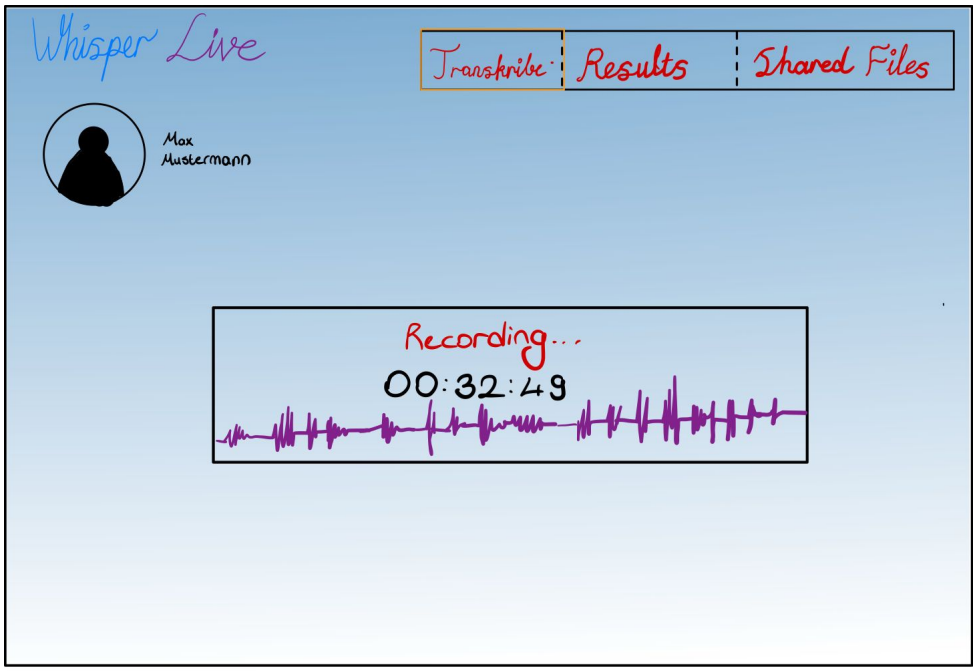


Figure 4: Transcribe View - Audio Recording State

5. Results View

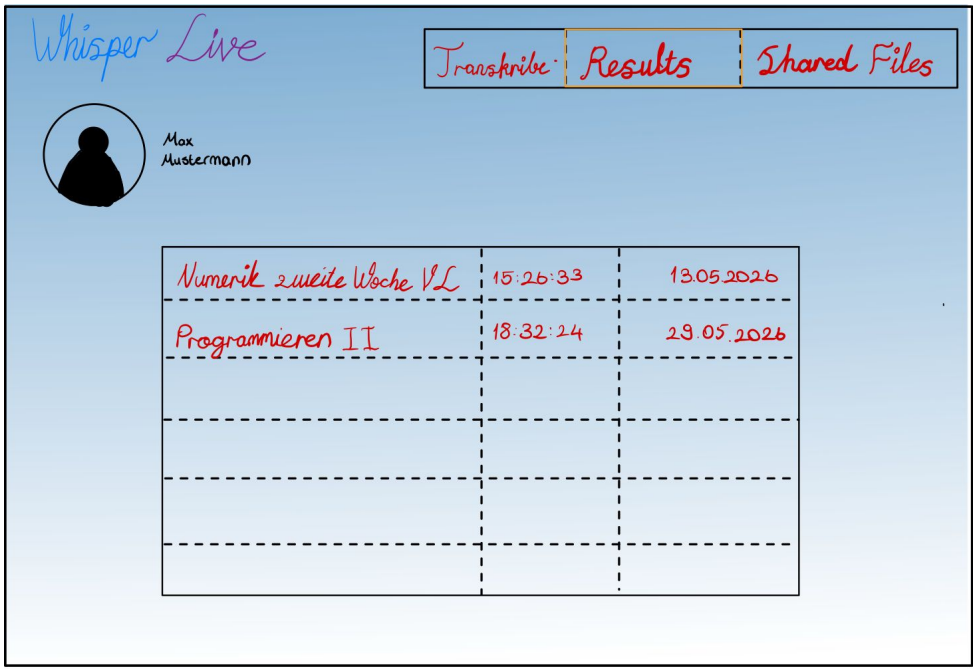


Figure 5: Results View Mockup

6. Shared Files View

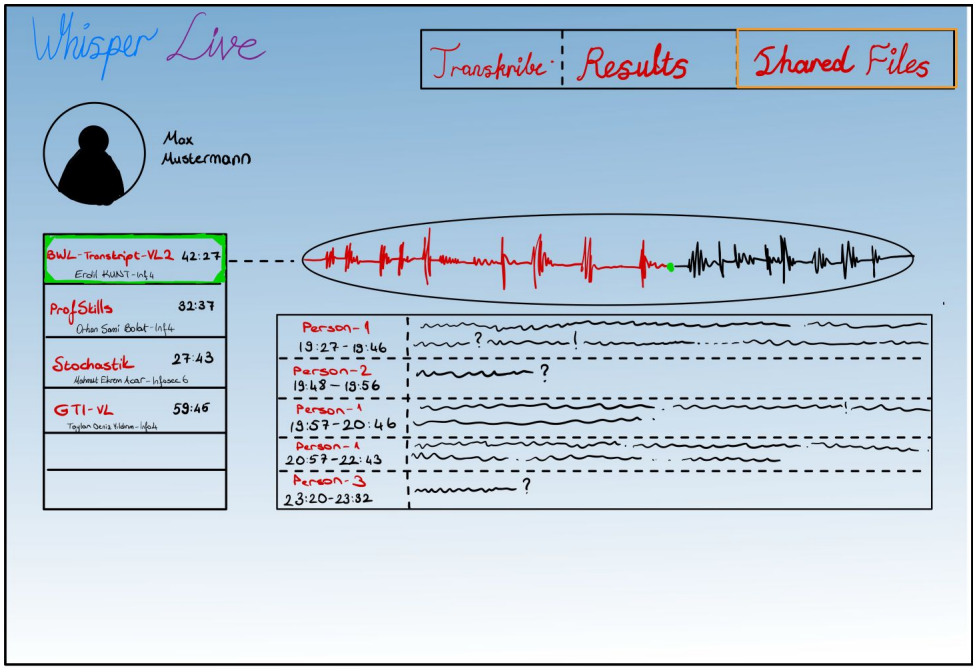
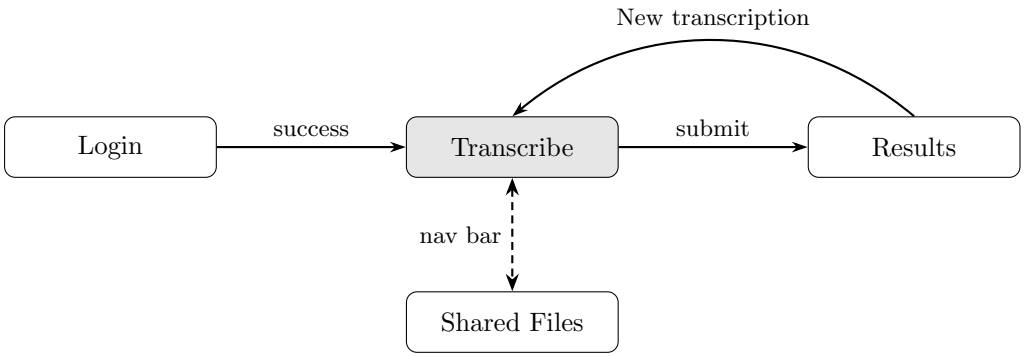


Figure 6: Shared Files View Mockup

Navigation and Screen Flow



The application begins at the Login view. Upon successful authentication, the user is automatically redirected to the Transcribe view. The Transcribe view serves as the central hub of the application: from here, the user is navigated to the Results view upon submitting an audio file, or they can access the Shared Files view via the navigation menu. The loading state is not a separate route; it is a conditional render within the Transcribe component while the `fetch` request is in flight.

Design Decisions

Authentication & Onboarding Flow

Context: The application serves a dual audience: incoming/outgoing exchange students and the general student body. The onboarding process must handle different credential types while securely routing users to their appropriate workspaces.

Decision 1: Bifurcated Login System

- **Unauthenticated State:** Users arriving at the homepage are prompted with an authentication modal.
- **Institutional Login (Primary):** Authenticates via university email (SSO/OAuth). This grants automatic access to both Personal Recordings and the shared Campus Hub for their specific institution (e.g., THWS networks).
- **Standard Login (Secondary):** Standard email/password for users who may not have their university credentials yet. This restricts access strictly to Personal Recordings until academic status is verified.

Workspace Routing (Post-Login)

Context: Once authenticated, users need immediate access to their study materials without navigating complex menus.

Decision 2: Dual-Hub Architecture

- Upon successful login, the system defaults to the Personal Dashboard (housing the user's private recording history and Calendar view).
- A persistent, top-level navigation toggle allows seamless switching between the Personal Dashboard and the Campus Hub (the shared repository for collaborative lecture transcripts).

Transcription Processing Architecture

Context: Transcribing audio requires significant computational power. We must offer flexible processing options to accommodate users with varying hardware capabilities and privacy requirements.

Decision 3: Hybrid Processing Model (Local vs. Remote)
The system will support two distinct transcription pipelines, configurable in the user's setup modal:

- **Pipeline A: Local Transcription (Privacy-First)**
  - **Architecture:** Utilizes a local instance of Ollama and an on-device Whisper Model.
  - **Workflow:** Uploaded/Recorded voice → Processed locally on the user's machine → UI chunks the output sentence-by-sentence for the post-lecture synthesis view.
  - **Trade-offs:** Ensures maximum data privacy and security for sensitive university lectures. However, it strictly requires the user to possess a suitably powerful CPU/GPU to prevent real-time lag.
- **Pipeline B: Remote/Token Transcription (Accessibility-First)**
  - **Architecture:** Cloud-based processing via the official Whisper API.
  - **Workflow:** User inputs their personal API token in account settings → Uploaded/Recorded voice is sent securely to the API → Output is returned and rendered in the exact same UI components as the local model.
  - **Trade-offs:** Eliminates hardware barriers, allowing any student with a basic laptop to use the tool. The trade-off is reliance on an external network connection and potential API token costs.

Responsiveness and Usability

Clean Layout & Workspace Navigation.
The interface is optimized for desktop environments, prioritizing a spacious, distraction-free experience for studying. A persistent, top-level navigation toggle allows students to instantly switch between their Personal Dashboard and the Campus Hub without navigating complex menus. The authentication and setup modals are centrally focused, ensuring an intuitive, clutter-free onboarding process. While designed primarily for desktop, the layout uses fluid containers to adapt gracefully to different window sizes, supporting students who use split-screen setups for note-taking.

Intuitive Input & Processing Feedback.
The application leverages standard desktop interaction patterns, featuring a prominent, easy-to-use drag-and-drop zone for audio file uploads alongside a direct recording option. Since audio transcription (especially via the local Pipeline A) can be computationally intensive, the UI provides clear, immediate visual progress indicators. This transparency manages user expectations, provides a smooth experience, and prevents accidental tab closures during processing.

Readability & Cognitive Load Reduction.
To assist students in reviewing dense lecture materials, the post-lecture synthesis view deliberately avoids overwhelming walls of text. The UI chunks the transcribed output sentence-by-sentence, significantly improving readability. Furthermore, generous whitespace is applied throughout the design to reduce cognitive fatigue. All interactive elements feature distinct

hover states to provide instant visual feedback for mouse users, and font sizes are defined in `rem` units to respect individual browser accessibility preferences.

AI Acknowledgement

This document was prepared with the assistance of AI tools (including Claude and Gemini) primarily for LaTeX formatting and language refinement. All core project concepts, design decisions, and final content were exclusively developed and reviewed by the student team.